

WEEK 2 TASK

Data Acquisition and Preliminary Analysis Execution

Project: Student Academic Performance Analysis

Dataset: UCI Student Performance – Mathematics Dataset (student-mat.csv)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

GitHub Project Repository

<https://github.com/Nupur125/data-exploration-strategy>

1. Introduction

The Week 2 task moves the project from planning to practical execution. The work focuses on acquiring a public dataset, checking its quality, preparing it for analysis, and performing preliminary exploratory analysis using Python. The project investigates student academic performance and factors that may be associated with final grades.

2. Dataset Identification

The UCI Student Performance dataset was selected. The Mathematics subset, `student-mat.csv`, contains 395 student records and 33 variables covering demographic information, family and education-related characteristics, study habits, absences, previous grades, and final academic performance. The final grade G3 is measured on a 0–20 scale.

3. Dataset Source and Justification

The dataset was obtained from the UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/320/student%2Bperformance>. It was selected because it is publicly available, documented, and contains numerical and categorical variables suitable for practical data-analysis work. Its variables allow investigation of study habits, absences, previous grades, internet access, and final performance.

4. Data Acquisition Strategy

The original CSV was stored in the GitHub repository under `data/raw/student-mat.csv`. Keeping the raw file unchanged preserves the original source data and supports reproducibility. The dataset can also be loaded directly from the GitHub raw URL in Python.

5. Python Tools and Methodology

Google Colab was used for execution. Pandas was used for data loading and manipulation, NumPy for numerical operations, Matplotlib for plotting, and Seaborn for statistical visualizations. The workflow consisted of acquisition, inspection, quality checks, cleaning, descriptive statistics, visualization, and preliminary interpretation.

6. Data Quality and Cleaning

The dataset was checked for structure, data types, missing values, duplicate records, descriptive statistics, and categorical consistency. A separate dataframe, `df_clean`, was created so the raw data remained unchanged. No observations were removed without a clear analytical reason. The cleaned dataset was saved separately as `student-mat-cleaned.csv`.

7. Preliminary Exploratory Analysis

Initial analysis examined final-grade distribution, study time versus final grade, absences versus final grade, previous grade G2 versus final grade G3, internet access versus final grade, and correlations among numerical variables. Histograms, bar plots, box plots, scatter plots, and a correlation heatmap were used.

8. Initial Findings

The analysis shows variation in final grades and provides several relationships for further investigation. Study-time groups can be compared using average G3. Absence counts can be examined against G3 to identify a possible association. The G2–G3 comparison helps assess whether previous academic performance is related to final performance. Internet access can be compared across grade distributions. These are preliminary observations and are not causal conclusions.

9. Reproducibility and GitHub

The project separates raw data, cleaned data, notebooks, reports, and supporting files. The raw dataset is stored in data/raw, the cleaned dataset in data/cleaned, and the analysis notebook in notebooks. The GitHub repository is: <https://github.com/Nupur125/data-exploration-strategy>

10. Conclusion

Week 2 successfully moved the project from strategy to execution. The public dataset was acquired, inspected, quality-checked, prepared, and explored using Python. The preliminary analysis provides a foundation for deeper statistical analysis and hypothesis testing in later stages.

11. Key Python Code Snippets

```
# Load dataset
url = "https://raw.githubusercontent.com/Nupur125/data-exploration-strategy/main/data/raw/student-mat.csv"
df = pd.read_csv(url, sep=";")

# Quality checks
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

# Cleaned copy
df_clean = df.copy()

# Descriptive statistics
df_clean.describe()

# Save cleaned data
df_clean.to_csv("student-mat-cleaned.csv", index=False)

# Correlation examples
df_clean["absences"].corr(df_clean["G3"])
df_clean["G2"].corr(df_clean["G3"])
```